

Late Surgical Complications Following Liver Transplantation

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Key Points

1. Biliary strictures and incisional hernias are the most common surgical complications encountered late after liver transplantation.
2. Anastomotic biliary strictures are amenable to endoscopic intervention and rarely need surgical intervention.
3. The presence of a biliary stricture mandates an evaluation of the patency of the hepatic artery.
4. Ischemic-type intrahepatic strictures are common indications for retransplantation.
5. Recipients of living related liver transplantation and donation after cardiac death allografts are at the highest risk for biliary and vascular complications late after transplantation. *Liver Transpl* 15:S12-S18, 2009. © 2009 AASLD.

Advances in surgical techniques, postoperative care, and immunosuppression over the last 20 years have significantly improved patient outcomes after liver transplantation. Nevertheless, this complex procedure remains associated with a variety of surgical complications that contribute to patient morbidity and mortality. Studies that have examined long-term outcomes after liver transplantation indicate that approximately 5% to 30% of recipients will develop vascular or biliary complications after transplantation.¹ Although these complications can occur both early (<3 months) and late (>3 months) after transplantation, there are often differences in the pathogenesis and presentation of these complications with time. Although early complications may result from imperfect surgical techniques and can be dramatic in presentation, late surgical complications are typically due to protracted ischemic or

immunogenic injury to the graft and often have a more indolent course.

In this work, we briefly review the presentation, etiology, therapy, and outcome of late surgical complications. We also discuss the influence of donor (ie, deceased versus living) and recipient (ie, adult versus pediatric) factors on the incidence of these complications.

SURGICAL COMPLICATIONS IN LIVER TRANSPLANTATION

General Surgical Complications

The most common general surgical complication encountered late after liver transplantation is herniation (Table 1). Incisional hernias are by far the most frequent hernia type, occurring in 5% to 25% of patients.² Risk factors implicated in the development of incisional hernias include obesity, sirolimus-based immunosuppression, and steroid boluses used to treat rejection. Incisional hernias can be observed in high-risk patients or can be successfully repaired with either open or laparoscopic herniorrhaphy in symptomatic patients; repair with a prosthetic mesh yields the lowest recurrence rate. Right-sided diaphragmatic hernias have also been observed in children following liver transplantation.³ Although the incidence of this complication is very low (<1%), surgical repair is recommended as these hernias appear to be more frequently associated with bowel obstruction.

Bowel obstruction is less common than herniation in liver transplant recipients (1% to 2%) but still remains an important source of morbidity. Bowel obstructions

Abbreviations: A2ALL, Adult-to-Adult Living Donor Liver Transplantation Cohort Study; BDFD, bile duct filling defect; CIT, cold ischemic time; DCD, donation after cardiac death; ERCP, endoscopic retrograde cholangiopancreatography; HAS, hepatic artery stenosis; HAT, hepatic artery thrombosis; ITBL, ischemic-type biliary lesion; LDLT, living donor liver transplantation; LRLT, living related liver transplantation; PTC, percutaneous transhepatic cholangiography; PVT, portal vein thrombosis.

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TABLE 1. Late Extrahepatic Surgical Complications for Adult Recipients of Whole-Organ Cadaveric Allografts

Type	Incidence	Etiology	Presentation	Therapy
Hernia	5%-18% (incisional)	Surgical technique Immunosuppression	Asymptomatic or a surgical emergency	No intervention or surgical herniorrhaphy
Bowel obstruction	1%-2%	Intraperitoneal surgical adhesions Internal hernias (Roux-en-y biliary- enteric anastomosis)	Abdominal pain Emesis Obstipation	Bowel rest and decompression Laparotomy and adhesiolysis

TABLE 2. Late Biliary Complications for Adult Recipients of Whole-Organ Cadaveric Allografts

Type	Incidence	Etiology	Presentation	Therapy
Biliary	5%-30%			
Anastomotic biliary stricture	0.6%-17.6%	Ischemia (HAT) Fibrotic healing Bile leak	Asymptomatic Cholangitis Graft dysfunction	Dilation and stenting (via PTC or ERCP) Biliary reconstruction (Roux-en-Y) if other measures fail
Nonanastomotic biliary stricture	5%-10%	HAT Immunogenic Prolonged CIT	Asymptomatic Cholangitis Biloma/abscess Graft dysfunction	Individualized: endoscopic therapy versus hepatic resection versus retransplantation
Biliary stones, sludge, and casts	5%	Biliary stricture Infection	Cholangitis Graft dysfunction	ERCP (preferred) or surgery with stone extraction

Abbreviations: CIT, cold ischemic time; ERCP, endoscopic retrograde cholangiopancreatography; HAT, hepatic artery thrombosis; PTC, percutaneous transhepatic cholangiography.

due to intraperitoneal adhesions, internal hernias, abdominal wall hernias (both incisional and inguinal), and neoplasms (including posttransplant lymphoproliferative disease) have all been reported.⁴ Notably, bowel obstructions in patients reconstructed with a Roux-en-Y choledochojejunostomy can be caused by herniation of a bowel segment through the mesenteric defect of the Roux limb. Thus, a high degree of suspicion for this specific complication must be maintained in patients who develop a bowel obstruction in the setting of a previous choledochojejunostomy. Notably, these patients can present with bowel strangulation in the absence of radiographic bowel obstruction; a contrast CT scan is often diagnostic of this life-threatening complication.

Biliary Complications

Strictures

Biliary strictures are the most frequent type of late biliary complication (Table 2) and are typically due to ischemia/reperfusion injury, vascular insufficiency, or fibrotic healing caused by improper technique. Importantly, hepatic artery stenosis (HAS) or hepatic artery thrombosis (HAT) may be present in 25% to 50% of patients with late biliary strictures.^{5,6} Hence, patients in whom either an anastomotic or nonanastomotic biliary stricture is identified should undergo an ultrasonographic or computed tomography evaluation of the he-

patic artery to determine patency. In addition, an angiographic study of the hepatic artery is useful in the evaluation of a recipient with a biliary stricture, as ultrasonography can be normal in patients who have developed HAT.

Nonanastomotic biliary strictures may develop in the absence of vascular insufficiency but still possess the cholangiographic appearance of ischemic strictures. Such strictures are called ischemic-type biliary lesions (ITBLs). Although ITBLs may affect either the intrahepatic or extrahepatic biliary tree, these lesions commonly affect the bifurcation of the bile duct (Fig. 1).⁶ The pathogenesis of these strictures remains unclear, but immunological injury to the bile duct epithelium has been proposed. Known risk factors include prolonged warm or cold ischemic time for the allograft, the use of highly viscous preservation solutions, and older donor age.⁶ As discussed further later, donation after cardiac death (DCD) allografts are also at higher risk for the development of ITBLs⁷ (Fig. 2). Most patients with ITBLs will present within 6 months of transplantation with fever and signs and/or symptoms of cholestasis. Cholangiography is the gold standard for diagnosis, although magnetic resonance cholangiopancreatography is being used with increasing frequency. Treatment of ITBL is difficult and is highly individualized. Although endoscopic dilation and stenting may successfully treat certain patients with proximal lesions,⁸ surgical therapy is frequently required in this population.⁹



Figure 1. Ischemic-type biliary lesions. Note the beading of the peripheral intrahepatic ducts and the stricture at the bifurcation in this endoscopic retrograde cholangiopancreatography image.

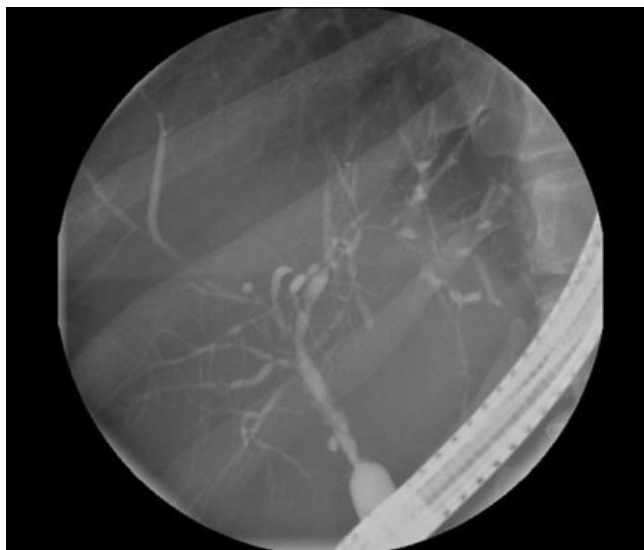


Figure 2. Ischemic-type biliary lesion in a donation after cardiac death allograft.

The decision to perform a Roux-en-Y reconstruction, hepatic resection, or retransplantation depends on the exact anatomy of the lesions as well as the symptomatology of the patient.

Anastomotic strictures are common after transplantation and can present up to several years after surgery. As with ITBLs, patients present with signs and/or symptoms of cholestasis such as elevated liver function tests, cholangitis, or even hepatic abscess. Magnetic resonance cholangiopancreatography or contrast cholangiography (either endoscopic retrograde cholangiography or percutaneous transhepatic cholangiography) is necessary to make the diagnosis (Fig. 3). Anastomotic strictures in patients reconstructed with a



Figure 3. Anastomotic biliary stricture. This endoscopic retrograde cholangiopancreatography image demonstrates a tight stricture at the choledochocholedochostomy.

choledochocholedochostomy are treated with endoscopic dilation and stenting with excellent outcomes, although multiple endoscopic sessions and stent replacement over several months may be required.^{8,10} Although endoscopic therapy is equally effective in patients with either early or late anastomotic strictures, patients with late anastomotic strictures are at a higher risk for stricture recurrence.¹⁰ Antegrade dilation and stenting can be performed via percutaneous transhepatic cholangiography in patients who possess a biliary-enteric anastomosis. Surgical management with Roux-en-Y biliary reconstruction is reserved for patients who fail endoscopic therapy. When they are treated successfully, long-term outcomes of patients with anastomotic strictures are equivalent to those of patients without anastomotic strictures.

Casts, Sludge, and Stones

Biliary stones, sludging, and casts may also develop late after transplantation. Cholangiographic evaluation of patients after liver transplantation suggests that 5% to 6% of patients will have bile duct filling defects (BDFDs), although the true incidence of BDFDs is unknown. Cholangitis is a common presentation of patients with biliary stones, sludging, or casts, and as many as 40% of patients with BDFDs have concomitant biliary strictures. Furthermore, Roux-en-Y biliary reconstruction may be a risk factor for the development of this disease, given the susceptibility of the bile duct to reflux cholangitis in the absence of a physiologic sphincter. Endoscopic therapy will successfully extract stones in up to 90% of patients, but patients with biliary casts often fail endoscopic therapy and may require surgical extraction, conversion to Roux-en-Y choledochojejunostomy, or retransplantation.^{8,11} Of historical note is the high degree of stone formation and biliary complications in patients who underwent cholecys-

TABLE 3. Late Vascular Complications for Adult Recipients of Whole-Organ Cadaveric Allografts

Type	Incidence	Etiology	Presentation	Therapy
Vascular	1%-5%			
Hepatic arterial thrombosis	1%-4%	Prothrombotic physiology Severe rejection	Asymptomatic (50%) Cholangitis/biliary strictures	Individualized: no therapy versus retransplantation
Hepatic artery aneurysm or pseudoaneurysm	1%	Arterial reconstruction Infected anastomosis Liver biopsy or intrahepatic stenting	Graft dysfunction Hypotension Gastrointestinal bleeding/hemobilia Biliary obstruction	Coil embolization versus surgical excision/ligation
Hepatic arterial stenosis	2%-4%	Fibrotic healing or Surgical technique	Asymptomatic or graft dysfunction	Angioplasty Retransplantation if there are biliary complications. Possible role for surgical hepatic artery revision
Portal vein thrombosis or stenosis	2%-3%	Anastomotic alteration Splenectomy	Asymptomatic Portal hypertension Ascites	Venoplasty/stent Surgical revision Retransplantation
Hepatic vein thrombosis or stenosis	1%-5%	Intimal hyperplasia	Extremity edema Ascites Dyspnea	Angioplasty Stenting Surgical revision

roduodenostomy or cholecystojejunostomy (i.e. the Waddell-Calne biliary reconstruction). Likewise, a high incidence of de novo cholelithiasis was seen in patients who retained donor gallbladders, presumably due to the denervated nature of the gallbladder, which necessitated late cholecystectomy.¹²⁻¹⁵

Vascular Complications

Aneurysms, thrombosis, or stenosis of the hepatic vasculature presents late after liver transplantation in fewer than 5% of patients (Table 3). Despite the low incidence of these complications, they are associated with substantial morbidity and may require retransplantation.

Among the late vascular complications seen in liver transplant recipients, HAT is the most common. Although early HAT may present in a fulminant manner, the presentation of late HAT is often very indolent. Approximately half of patients who present with late HAT will be asymptomatic,¹⁶ and the thrombosis will be discovered during routine ultrasonographic evaluation of the hepatic vasculature. These patients can be managed conservatively but expectantly, as 25% to 30% will develop biliary strictures or graft failure and require retransplantation. Symptomatic patients with delayed HAT almost universally present with biliary symptoms such as a bile leak, an abscess, a stricture, or recurrent cholangitis. Although these patients can be initially treated with biliary stents or with biliary or vascular reconstruction, virtually all of these patients ultimately require retransplantation. Importantly, the window of opportunity in which these patients can be retransplanted is short, as biliary sepsis is common and contributes significantly to the 50% mortality rate observed with retransplantation.

HAS may develop early but is usually diagnosed >3

months after transplantation.¹⁷ Although the true incidence of HAS is unknown as many patients are asymptomatic, the most common clinical presentation of this anomaly is abnormal liver function tests. Biliary complications as a result of HAS are less frequent than with HAT, as vascular collaterals often provide adequate blood flow to the allograft. The goal of treatment is to prevent future biliary complications, and balloon angioplasty is often effective. However, the benefit of HAS therapy has yet to be firmly established as graft and patient survival for patients treated for HAS and patients managed conservatively for HAS is equivalent.¹⁷

Hepatic artery aneurysms or pseudoaneurysms are rare complications of liver transplantation but are associated with high mortality. These vascular complications most often present early after transplantation but can present late. Extrahepatic arterial aneurysms at the anastomosis are almost always associated with fungal infection¹⁸ and are infrequently diagnosed radiographically. Although some aneurysms can cause compression and obstruction of the biliary tree, most hepatic arterial aneurysms present with rupture and hypotension, and patients may bleed intraperitoneally or into the gastrointestinal tract (ie, hemobilia). Intrahepatic arterial aneurysms and pseudoaneurysms are usually iatrogenic in nature, following liver biopsy or stent placement. Intrahepatic aneurysms or pseudoaneurysms can frequently be treated with coil embolization (Fig. 4), but patients presenting with hemorrhagic shock are best managed in the operating room with excision or ligation of the hepatic artery and subsequent arterial reconstruction. Despite these therapeutic measures, hepatic artery aneurysms are fatal in >50% of cases.

Portal vein stenosis and portal vein thrombosis (PVT) are also rare late complications of liver transplantation

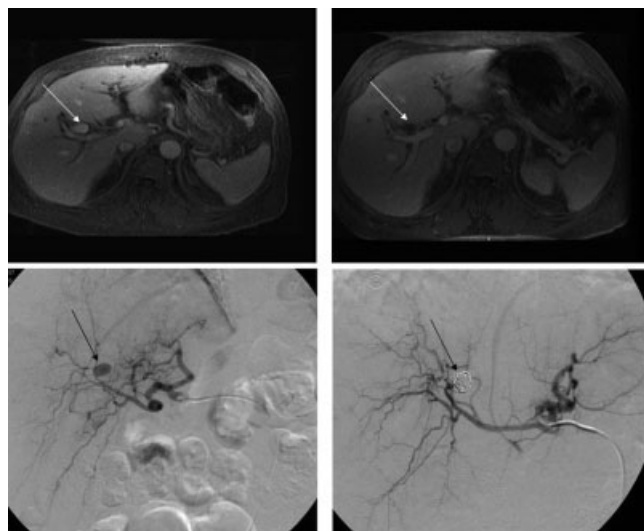


Figure 4. Intrahepatic arterial aneurysm. Left: Arrows delineate an aneurysm of the intrahepatic arterial tree causing hemobilia in a liver transplant recipient [magnetic resonance imaging (top) and angiogram (bottom)]. Right: Exclusion of flow in the aneurysm after successful coil embolization [magnetic resonance imaging (top) and angiogram (bottom)].

(1%-2%). Although symptoms associated with portal vein stenosis are unusual, stenosis greater than 80% or PVT may result in recurrent portal hypertension and ascites. As with HAT, most PVT occurs early after transplantation, but 30% of PVT may occur late. In important contrast to HAT, however, PVT negatively affects long-term graft and patient survival.¹⁹ This is likely due to the fact that retransplantation is less frequently an option after PVT because of the lack of appropriate venous inflow from the mesenteric circulation in patients with severe PVT.

Therapy for portal vein stenosis or thrombosis is tailored to symptomatic patients. Portal venous angioplasty with stenting currently yields excellent results and is the treatment of choice for symptomatic portal vein stenosis.²⁰ The role of percutaneous treatment for PVT, however, is still evolving. To date, only small case series of portal vein thrombolysis as well as percutaneous portal vein thrombectomy and stenting have been reported in the literature.²¹ Surgical mesenteric-intrahepatic portal vein shunting using an autogenous vein (ie, Rex shunt) has been reported to successfully treat PVT in liver transplant recipients.²² Nevertheless, percutaneous approaches may prove superior to surgical intervention with additional experience in the future.

Finally, hepatic vein or inferior vena cava stenosis and/or thrombosis can be a late vascular complication of liver transplantation as well. As with the other vascular complications reported herein, this complication is rare. Presenting symptoms are related to the exact anatomy of the obstruction and may include graft failure, lower extremity edema, signs/symptoms of portal hypertension, and refractory ascites. Imperfect surgical technique contributes to approximately one-third of cases (particularly when this complication presents

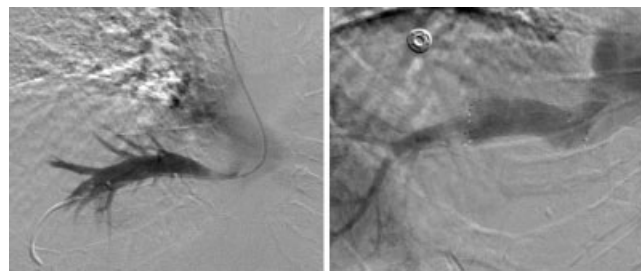


Figure 5. Hepatic vein stenosis. Left: A contrast venogram demonstrates stenosis of the right hepatic vein in a liver transplant recipient who developed liver dysfunction. Right: An expandable stent placed percutaneously via the internal jugular vein improved hepatic outflow into the inferior vena cava.

early); however, the recurrence of Budd-Chiari syndrome may also contribute to venous outflow thrombosis or stenosis. Percutaneous angioplasty via the internal jugular vein has excellent results with improvement of symptoms in over 80% of patients. However, the durability of these results is not as robust, and most patients require multiple sessions in order to maintain patency.²³ Nevertheless, complications are rare, and percutaneous therapy has replaced open surgical therapy in the vast majority of cases (Fig. 5).

INFLUENCE OF THE DONOR TYPE ON LATE COMPLICATIONS

The severe organ shortage has prompted many US centers to adopt strategies historically employed in Asia and expand their recovery programs to include donors outside the standard whole-organ cadaveric pool. However, factors unique to these donors may affect long-term outcomes after liver transplantation, including the development of late complications.

DCD

No studies exist to date that have discretely evaluated late surgical complications in recipients of DCD allografts. However, surgical complications in these recipients may be more frequent, given the increased ischemic time associated with the procurement of these allografts. Indeed, several studies that have evaluated vascular and biliary complications in DCD recipients have suggested that these allografts are at higher risk for biliary complications.⁷ Specifically, DCD livers have an elevated risk for developing nonanastomotic biliary strictures or ITBLs (Fig. 2), although there appears to be no increased risk for the development of anastomotic strictures or vascular complications. The increased frequency of ITBLs in DCD livers provides additional evidence that increased ischemic time, particularly warm ischemic time, promotes the development of biliary strictures. Furthermore, the increased risk of ITBLs in DCD livers is consistent with both the known susceptibility of biliary epithelium to ischemia-reperfusion injury and the hypothesized role of ischemia-reperfusion injury in the formation of ITBLs.

Living Donors

Multiple groups from Asian centers with experience in living donor liver transplantation (LDLT) have established an increased incidence of surgical complications in recipients of these allografts.²⁴ Recent data from the ongoing multicenter Adult-To-Adult Living Donor Liver Transplantation Cohort Study (A2ALL) in the United States further support these findings and indicate that recipients of a right lobe allograft from an adult living donor are at elevated risk for both biliary and vascular complications (specifically bile leaks, HAT, and PVT).²⁵ Unfortunately, the published data do not specify the precise onset of these complications, and follow-up at this time is somewhat limited. It thus remains unclear whether there is an increased incidence of late complications in LDLT recipients compared to whole-organ recipients. Although donor-recipient size discrepancies may be greater in LDLT recipients and potentially increase anastomotic stricture incidence, it is possible that decreased ischemic time may decrease the incidence of ITBLs that develop later after transplantation.

LATE SURGICAL COMPLICATIONS AFTER PEDIATRIC LIVER TRANSPLANTATION

The frequent utilization of reduced size or split allografts [including living related liver transplantation (LRLT)] and the common need for vascular reconstruction in pediatric recipients contribute to an increased incidence of both biliary and vascular complications in pediatric recipients. These late complications are most often biliary and venous in nature, as the wide application of microsurgical techniques and routine anticoagulation or antiplatelet therapy in pediatric recipients has significantly decreased the historically high rate of HAT in this population. Modern case series report that late anastomotic and nonanastomotic biliary strictures occur in approximately 7% of pediatric recipients.²⁶ Importantly, however, a higher stricture rate of 24% has been reported in recipients of LRLT²⁷ because of the reconstructive challenges often associated with the variable biliary anatomy of donor segments II to IV, which are commonly transplanted into children. Similarly, reconstruction of the hepatic outflow or portal vein in pediatric recipients of LRLT may result in late venous stenosis in almost one-third of patients.²⁸

SUMMARY

Surgical complications after liver transplantation are most frequent within the first 90 days but do occur late after surgery. Common late complications include incisional hernias as well as biliary strictures, although rarer complications (HAT and PVT) can result in substantial morbidity. Percutaneous therapies (ie, balloon angioplasty and stenting) provided by an experienced interventional radiologist are commonly employed and have supplanted surgery as the therapy of choice in all but the most intractable cases. Physicians caring for liver transplant recipients must be aware that certain

donor allografts (such as those from DCD or living donors) may increase the risk of late surgical complications.

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